

Entities don't receive knockback from projectiles fired from dispensers

MC-44280 Projectiles

Status	Resolved / Fixed
Affected	Minecraft 1.7.4, Minecraft 1.7.9, Minecraft 1.8, Minecraft 1.8.3, Minecraft 15w47c, Minecraft 1.9 Pre-Release 2 (+59 more)
Fixed in	Minecraft 16w02a, 24w18a
Reported	2014-01-07

STEPS TO REPRODUCE

1. Build the setup shown in `setup.png`.
2. Place snowballs inside the dispenser.
3. Summon a husk on the diamond block.
4. Use the lever to activate the dispenser.
5. Observe that the snowballs hit the husk without knocking it back.

OBSERVED BEHAVIOUR

When a dispenser launches a snowball, egg, firework rocket, or splash potion of harming at an entity, the projectile can hit the entity but applies no knockback. The affected entity remains in place, unlike when the same projectile is thrown by a player or launched by a mob such as a snow golem or witch.

Arrows are not affected: arrows fired from dispensers apply knockback normally, as they do when fired from a bow.

EXPECTED BEHAVIOUR

When a dispenser fires a snowball at an entity, the projectile should apply knockback on impact, causing the husk to be pushed away from the dispenser. Dispenser-fired projectiles such as snowballs, eggs, fireworks, and splash potions of harming should produce the same knockback behavior as when launched by a player or an entity, rather than dealing their effects without moving the target.

ENVIRONMENT

Minecraft Java Edition Versions: 1.7.4, 1.7.9, 1.8, 1.8.3, 15w47c, 1.9 Pre-Release 2, 1.9.2, 1.9.3 Pre-Release 1, 1.9.3 Pre-Release 2, 1.9.4, 16w20a, 1.10.2, 16w32a, 16w43a, 1.11.2, 17w06a, 17w15a, 1.12.2, 17w45b, 17w46a, 17w47a, 17w47b, 18w02a, 18w11a, 18w16a, 18w20c, 18w21a, 1.13.1, 1.13.2, 18w48a, 18w48b, 18w49a, 19w04b, 19w05a, 1.14 Pre-Release 5, 1.14.4, 19w36a, 19w37a, 20w06a, 1.16.2, 1.16.3, 1.16.4 Pre-release 2, 1.16.4, 20w46a, 20w51a, 21w05b, 21w06a, 21w07a, 21w11a, 21w14a, 21w17a, 1.17, 1.17.1, 21w39a, 1.18.1, 1.18.2, 22w17a, 1.19, 1.19.2, 1.19.3 Release Candidate 3, 1.19.3, 1.20, 1.20.1, 1.20.2, 1.20.4 (includes snapshots)

ORIGINAL DESCRIPTION

Fireworks, eggs, snowballs, and splash potions of harming fired from dispensers don't knock back mobs, but they do when thrown by the player or a mob such as a snow golem or witch.

Steps to Reproduce:

- Build the setup as shown in the attachment below. setup.png
- Place some snowballs inside of the dispenser.
- Summon a husk on the diamond block.
- Use the lever to activate the dispenser and watch the husk closely.

Observed Behavior:

Projectiles fired from dispensers don't deal any knockback to entities.

Expected Behavior:

Projectiles fired from dispensers would deal knockback to entities.

Note

Arrows are not affected by this bug, as they knock back mobs when fired either from a bow or a dispenser.

Code analysis

Code analysis by marcono1234 can be found in this comment.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-44280>